

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary Summer 2024 Course: B. Tech. Semester : VI Branch : Mechanical/automobile/Production/Automation & Robotics/ Mechatronics Subject Code & Name: BTMPE604D Robotics Max Marks: 60 Date: 21/06/2024 Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the role of following components in robot operations (i) manipulator (ii) Controller (iii) End Effector.	CO1	6
B)	Describe any three robot configurations. Draw neat sketch and write applications.	CO1	6
C)	Obtain homogeneous transformation matrix.	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Explain different types of links and joints which are used in robotics.	CO2	6
B)	Write criteria selection for electrical drives.	CO3	6
C)	What is forward and reverse kinematics? Explain.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	What are the key features of hydraulics drives? Write limitations of the same.	CO3	6
B)	Discuss working and applications of servo motors .	CO3	6
C)	Compare Proximity Sensors: Inductive, Capacitive, and Photoelectric	CO4	6
Q.4	Solve Any Two of the following.		12
A)	What are different types of end effectors? Explain any two in detail.	CO4	6
B)	Write basic considerations that are used in gripper selection and design.	CO4	6
C)	Write the origin and applications of following programming languages AL,AML,RAIL	CO5	6
Q. 5	Solve Any Two of the following.		12
A)	Discuss in brief; economic analysis for robotic implementation.	CO5	6
B)	With neat sketch explain any one application using robot in quality inspection.	CO5	6
C)	Explain following robot programming methods (i) Lead through (ii) Teach Pendant	CO5	6
	*** End ***		